
Time Difference of Arrival Localization: Principles, Geometry, and the Complete Analytical Solution

A rigorous tutorial introduction to TDoA emitter localization, centered on the closed-form method of Hubáček, Veselý & Olivová (Defence Science Journal, 2022)

TECHNICAL EXPLAINER · JUNE 2026 · FOR LEARNERS ENTERING THE TDOA FIELD

ABOUT THIS DOCUMENT. Time Difference of Arrival (TDoA) localization fixes the position of a transmitter that does not cooperate with the measuring system: the receivers never learn when the signal was sent, only when it arrived at each of them. This explainer develops the method from first principles — the hyperbolic geometry of constant range difference, the sensor-counting rules, a survey of the classical solution algorithms — and then works through, step by step, the *complete analytical solution* of Hubáček, Veselý and Olivová [1]: the coordinate-transformation trick, the quadratic whose two roots are the candidate target ranges, root classification, the solvability and ambiguity structure of the plane, and the covariance-weighted extension to overdetermined networks. Error analysis (GDOP, error ellipses, the Cramér–Rao bound), the signal-processing front end (generalized cross-correlation), and practical engineering considerations complete the picture. All numerical examples in Section 6 were verified independently by direct computation, and all figures are drawn from exact mathematics, not artistic sketches.

1. Introduction and Applications

Most everyday positioning is *cooperative*: a GPS receiver knows the satellite signals and their transmission times; a radar clocks the round trip of its own pulse. The harder problem is locating a **non-cooperative emitter** — a transmitter that does not announce when it transmitted, what it transmitted, or that it is willing to be found at all. A hostile radar, an unlicensed broadcast, a phone dialing an emergency number, a gunshot, a calling whale: the signal simply arrives, and the only universally available observable is *when* it arrives at each of several receivers. Time Difference of Arrival (TDoA) localization is built on exactly that observable. Because the transmission instant t_0 is unknown, no single receiver can convert its arrival time into a range; but the *difference* between arrival times at two synchronized receivers does not depend on t_0 at all, and it confines the emitter to a well-defined curve. With enough receivers, the curves intersect at the emitter.

The idea has a long pedigree. Hyperbolic navigation systems of the LORAN family (and Decca and Omega) used it in reverse from the 1940s onward: chains of time-synchronized *transmitters* broadcast coordinated pulses, and a ship measured received time differences to place itself on intersecting hyperbolae printed on charts. Modern satellite navigation is a close cousin: GNSS pseudoranges all share one unknown receiver clock bias — the exact analogue of the unknown t_0 — and algebraic solutions such as Bancroft's [4] handle that shared unknown in closed form. Running the geometry forward — known receivers, unknown emitter — gives the applications that motivate this document:

- **Emergency call location (E-911).** Regulatory mandates for locating wireless emergency callers pushed network-based techniques such as uplink TDoA (U-TDOA): the handset's transmission is time-stamped at several base stations and the caller is multilaterated with no modification to the phone.
- **Gunshot detection.** Urban acoustic sensor networks and counter-sniper systems detect the muzzle blast at distributed microphones and localize the shooter from arrival-time differences within fractions of a second.

- **Acoustic source localization.** Microphone arrays steer cameras toward talkers and localize sound sources in robotics; the cross-correlation front end of Section 9 was developed largely in this context [10].
- **Wildlife tracking.** Grids of autonomous recorders — and hydrophone arrays for whales — turn arrival-time differences of animal calls into positions and movement tracks without tagging the animal.
- **Electronic warfare and ELINT.** Passive surveillance systems locate radars and other emitters without radiating a single watt, so they cannot be detected by anti-radiation means and do not betray their own position. This is the context of the paper at the heart of this explainer [1], written at the University of Defence in Brno; the Czech school of passive surveillance (the Tamara and VERA families of deployed systems) is a well-known TDoA lineage, and airport multilateration of transponder signals is its civil sibling.

All of these share the problem statement that defines TDoA: *given N time-synchronized sensors at known positions, each recording the arrival time of the same signal, estimate the emitter position when the emission time is unknown.* The document develops the solution in order: measurement model (Section 2), hyperbolic geometry (Section 3), sensor counting (Section 4), the landscape of solution algorithms (Section 5), the complete analytical solution in full detail (Section 6), its two-fold ambiguity (Section 7), error statistics (Section 8), estimating the time differences themselves (Section 9), and the engineering realities of fielded systems (Section 10).

2. The Principle: Differencing Away the Unknown Transmit Time

Let the emitter (the “target” in ELINT parlance) sit at unknown position $\mathbf{x}_t$ and transmit at unknown time t_0 . Sensor i sits at known position $\mathbf{s}_i$, and the signal propagates at speed c (the speed of light for radio, the speed of sound for acoustics). Under line-of-sight propagation the time of arrival measured by sensor i is

$$\text{TOA}_i = t_0 + \frac{r_i}{c}, \quad r_i = \|\mathbf{x}_t - \mathbf{s}_i\|.$$

Each equation contains the same nuisance unknown t_0 . One measurement therefore tells us nothing about range by itself. But subtracting the equation for a reference sensor (say sensor 1) from the equation for sensor i eliminates t_0 exactly:

$$\Delta t_{i1} = \text{TOA}_i - \text{TOA}_1 = \frac{r_i - r_1}{c} \implies \Delta r_{i1} = c \Delta t_{i1} = r_i - r_1.$$

The measured time difference Δt_{i1} , scaled by the propagation speed, is a *range difference*: the difference between the (unknown) distances from the emitter to two known sensors. This is the fundamental TDoA observable. Two remarks deserve emphasis. First, the differencing only works if the sensors share a common time base — inter-sensor synchronization is a prerequisite whose quality directly limits accuracy (Section 10); the emitter, by contrast, needs no synchronization at all, since its clock has been differenced out. Second, a range difference is a much weaker constraint than a range: a known range puts the emitter on a circle (sphere in 3D), while a range *difference* puts it on an unbounded hyperbola branch (hyperboloid sheet in 3D). That weakening is the price of non-cooperation, and the geometry of Section 3 is its consequence.

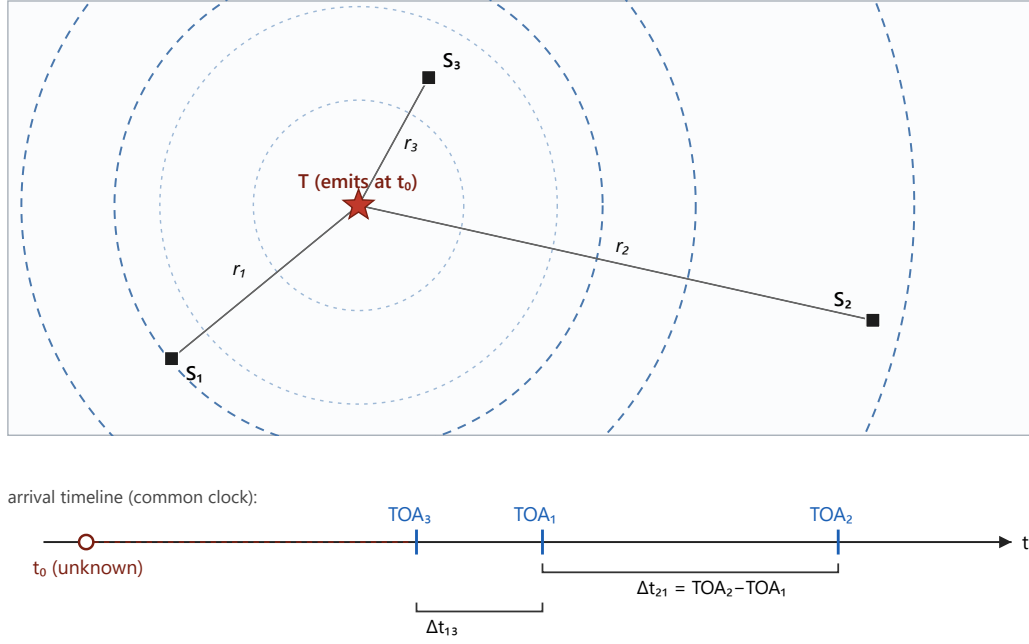


Figure 1. The TDoA principle. *Top:* an emitter T transmits once, at unknown time t_0 ; the circular wavefront (dashed circles, drawn to true scale through each sensor) reaches sensor S_3 first, then S_1 , then S_2 , because the ranges satisfy $r_3 < r_1 < r_2$. *Bottom:* the arrival timeline. The absolute offset from t_0 to the arrivals is unknown (the whole tick pattern can slide rigidly), but the differences Δt_{21} and Δt_{31} are measurable and each converts to a range difference $\Delta r = c \Delta t$.

3. Geometry: One Branch of a Hyperbola

Fix two sensors s_i and s_j separated by baseline $d = \|s_i - s_j\|$, and suppose the measurement gives $\Delta r = r_i - r_j$. The locus of points whose distances to two fixed points differ by a constant is, by the classical definition, a **hyperbola with foci at the two sensors** — but only *one branch* of it. A full hyperbola is the set $|r_i - r_j| = \text{const}$; the signed measurement $r_i - r_j = \Delta r$ selects the branch that wraps around the *nearer* sensor, i.e. the one the signal reached first. If $\Delta r > 0$ (arrival at j earlier), the emitter is closer to s_j and lies on the branch curving around s_j ; if $\Delta r < 0$, around s_i . In the canonical frame with the foci on the x -axis at $\pm c_f$, the branch parameters are

$$a = \frac{|\Delta r|}{2}, \quad c_f = \frac{d}{2}, \quad b^2 = c_f^2 - a^2, \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

where a is the semi-major (semi-transverse) axis, c_f the focal half-distance, and b the semi-minor (semi-conjugate) axis. As the curve recedes from the sensors it straightens onto its asymptotes $y = \pm(b/a)x$ — a fact with major consequences for far-field accuracy (Section 8).

The triangle inequality bounds the observable: no point in the plane can have $|r_i - r_j| > d$, so a valid (noise-free) measurement always satisfies $|\Delta r| < d$; with measurement noise an apparent $|\Delta r| \geq d$ can occur and must be rejected or clamped, since b^2 above would become non-positive. Two degenerate cases bracket the family. When $\Delta r = 0$ the “hyperbola” flattens into the **perpendicular bisector** of the baseline: the emitter is equidistant from both sensors. When $|\Delta r| = d$ (possible only for emitters exactly on the baseline’s extension) the branch collapses to a **ray** along the line through the sensors, starting at the nearer sensor and pointing away from the other. Between these extremes, smaller $|\Delta r|$ gives an open, nearly straight curve; $|\Delta r|$ close to d gives a tight, sharply bent branch hugging the near sensor.

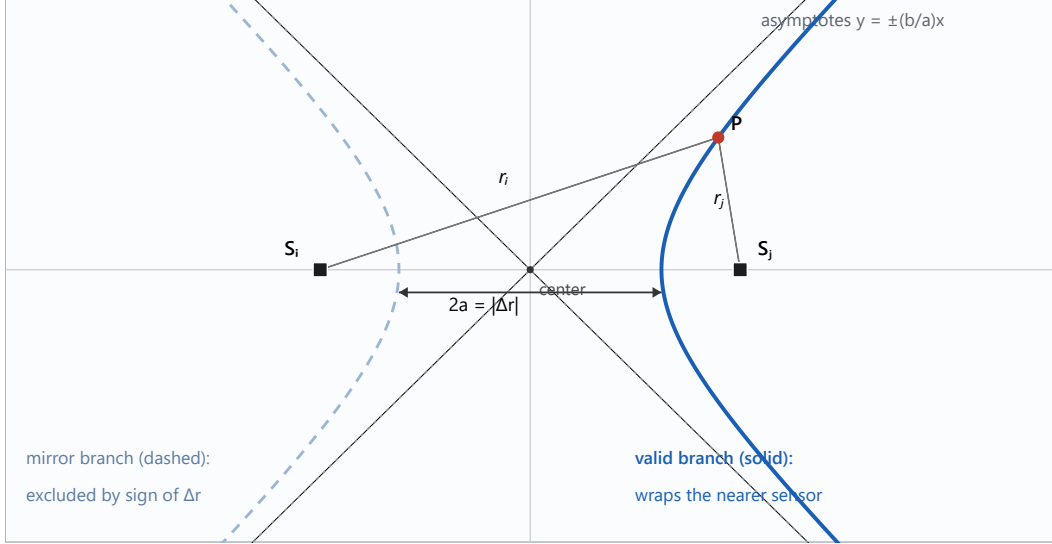


Figure 2. The locus of constant range difference for one sensor pair (foci S_i, S_j ; baseline $d = 8$, $\Delta r = r_i - r_j = +5$, hence $a = 2.5$, $c_f = 4$, $b = \sqrt{c_f^2 - a^2} \approx 3.122$). The measured sign selects the solid branch wrapping the nearer sensor S_j ; the dashed mirror branch corresponds to $\Delta r = -5$ and is excluded by the data. Dotted lines are the asymptotes $y = \pm(b/a)x$: far from the array, the usable locus degenerates toward a pure direction. The marked point P illustrates membership: $r_i - r_j = 2a$ for every point of the solid branch.

3.1 Forming the TDOA set: centralized vs. decentralized

With N sensors there are two common ways to organize the differences. In the **centralized** (common-reference) formation, one station — the central or reference station, often the site hosting the processing — is subtracted from all others, producing the $N - 1$ observables $\Delta t_{21}, \dots, \Delta t_{N1}$. In the **decentralized** (pairwise) formation, every pair is differenced, producing $N(N - 1)/2$ values. The pairwise set looks richer but is not: every pairwise difference is a linear combination of the common-reference ones, $\Delta t_{ij} = \Delta t_{i1} - \Delta t_{j1}$, so **only $N - 1$ of the $N(N - 1)/2$ pairwise TDOAs are independent**. A properly weighted estimator using the $N - 1$ common-reference TDOAs (with the correlated covariance of Section 8) extracts everything the full set contains; the real distinction is architectural — what is time-stamped where — rather than informational. The method of Section 6 uses the centralized formation with S_1 as reference.

4. Multilateration and Determinedness

Locating an emitter from several range differences is called **hyperbolic multilateration**. The counting rule follows from Section 3: in n dimensions the position has n unknowns, and N sensors yield $N - 1$ independent TDOAs, so the system is exactly (“sufficiently”) determined when $N - 1 = n$, i.e. with $n + 1$ **sensors: three in 2D, four in 3D** (the paper [1] states this as “ $N + 1$ sensors in N -dimensional space”). Geometrically, each TDOA confines the emitter to one hyperbola branch (2D) or one sheet of a hyperboloid of two sheets (3D), and the position estimate is the **intersection** of those curves or surfaces: two branches meeting in a point (or two — Section 7) in the plane, three sheets meeting in space.

Let it be stated clearly: **with more than $n + 1$ sensors the system is overdetermined — beyond 4 nodes for 3D localization, beyond 3 for 2D** — the independent equations then outnumber the unknowns. Overdetermination is not a defect; fielded systems seek it deliberately: redundancy averages down noise, extends coverage, exposes faulty sensors, and (Section 7) resolves the two-fold ambiguity of exactly determined TDoA. But it changes the mathematics: with noisy measurements the hyperbolae no longer intersect in a single point, and “the solution” must be defined statistically — by least squares, maximum likelihood, or, as in Section 6.9, covariance-weighted fusion of closed-form sub-solutions. Nor is

the exactly determined case trivial: the equations are quadratic, so even three perfect 2D measurements can be consistent with two distinct emitter positions.

5. A Survey of Solution Methods

The TDoA equations are nonlinear, and half a century of literature offers two broad attacks: iterate from a guess, or algebraically restructure the equations until they surrender a closed form. The following seven methods are the classical landmarks; the method of Section 6 belongs to the closed-form family and should be read against this background.

(a) Iterative Taylor-series / Gauss–Newton linearization (Foy, 1976) [2]. Foy’s method expands the range-difference equations in a first-order Taylor series about an initial position guess, solves the linear least-squares problem for a correction, and repeats until the correction is negligible. It is flexible (any number of sensors, easy weighting, natural covariance output) and asymptotically optimal when it converges to the right place — but it *requires an initial guess*, and the iteration can stall, diverge, or converge to the wrong solution. The failure mode is not hypothetical: near the borders of the ambiguous areas analyzed in Sections 6–7, the true and ghost solutions lie in adjacent basins of attraction, and a mediocre guess sends Gauss–Newton confidently to the ghost. Hubáček et al. [1] cite exactly this behavior as a principal motivation for an analytical alternative whose every solution is enumerated and classified rather than stumbled upon.

(b) Spherical-interpolation least squares (Smith & Abel, 1987) [3]. Smith and Abel observed that if the unknown range from a reference sensor, $R_s = r_1$, is treated for a moment as an independent variable, the squared range-difference equations become *linear* in the emitter coordinates and R_s jointly. Their spherical-interpolation estimator solves this augmented linear system in the least-squares sense, then eliminates the auxiliary variable by orthogonal projection — a non-iterative closed form for any number of sensors. Because it minimizes an *equation error* rather than the measurement error it is not statistically efficient, and accuracy degrades for distant sources; but it is robust, cheap, free of convergence concerns, and remains the standard warm start for iterative refiners. The auxiliary-range idea is the seed from which both Chan–Ho and the Section 6 method grew.

(c) Bancroft’s algebraic GPS solution (1985) [4]. Bancroft solved the closely related TOA/pseudorange problem — GPS equations in which all measured ranges share one unknown receiver clock bias — in closed form using a Lorentzian (Minkowski) inner product on position–time 4-vectors. The shared clock bias plays exactly the role of the emitter’s unknown t_0 in TDoA, which is why the method belongs in this survey: differencing pseudoranges converts Bancroft’s problem into a TDoA problem and vice versa. The algebra collapses to a single scalar quadratic whose *two roots* generate two candidate solutions — the same two-fold ambiguity that Section 6 meets as the quadratic in K . For satellite geometry one root is easily discarded (it lies far from the Earth’s surface); for terrestrial TDoA the discard is not always possible, which is what makes ambiguity analysis necessary.

(d) Chan & Ho’s two-step weighted least squares (1994) [5]. The estimator of Chan and Ho (*IEEE Trans. Signal Processing* **42**(8), pp. 1905–1915) is the reference algorithm of the field. Step one pseudo-linearizes the equations exactly as in (b) — the auxiliary variable r_1 is appended to the unknowns — but solves them by *weighted* least squares with weights derived from the actual TDOA noise covariance. Step two exploits the dependence step one ignored: the components of the step-one estimate are squared and combined in a second WLS problem that enforces the constraint $r_1^2 = \|\mathbf{x}_t - \mathbf{s}_1\|^2$, tightening the estimate. The procedure is non-iterative, handles arbitrary N , and is provably *asymptotically efficient*: as TDOA noise becomes small its covariance attains the Cramér–Rao bound. Its weaknesses are a noise-threshold effect from the squaring step and degradation very near or very far from the array; numerous variants patch these cases.

(e) Fang’s exact solution (1990) [6]. Fang (*IEEE Trans. Aerospace and Electronic Systems* **26**(5)) gave the canonical exact solution for the *exactly determined* case — three sensors in 2D, four in 3D. The key move is a coordinate frame adapted to the sensors: reference at the origin, second sensor on the x -axis, so each

squared difference equation loses its quadratic terms and the system reduces to one quadratic in one variable. Fang’s solution is exact and iteration-free, but it consumes exactly as many measurements as there are unknowns — redundant sensors cannot be used — and it inherits the two-root ambiguity undiluted. The reader will recognize Fang’s coordinate adaptation as the same device that powers the method of Section 6; what [1] adds is the complete classification of roots, solvability, accuracy, and the fusion machinery for overdetermined networks.

(f) The Bard–Ham–Jones algebraic solution (1996) [7]. Bard, Ham and Jones (SOUTHEASTCON ’96) formulated the TDoA equation set in matrix form with the reference range as auxiliary unknown and solved it algebraically, giving explicit conditions under which the solution is unique, ambiguous, or nonexistent. The formulation exposes the algebraic structure of the problem well; computationally it represents the *matrix-operation family* of analytical solutions — those that form and invert (or pseudo-invert) matrices per fix. This family is the benchmark against which the direct algorithm of Section 6 was measured in the complexity study [13]: the direct algorithm avoids matrix inversion altogether (Section 6.9).

(g) The Mellen–Pachter–Raquet closed form (2003) [8]. Mellen, Pachter and Raquet derived a compact exact closed-form TDoA solution (MPR) that, like (b) and (d), passes through the auxiliary reference range: the linearized system is solved for position as an affine function of r_1 , substitution into the defining quadratic for r_1 yields a scalar quadratic, and back-substitution finishes. Its contributions are economy and a clean handling of the overdetermined case, where the linear solve becomes a pseudoinverse solve while the scalar quadratic structure survives. MPR demonstrates that “affine in range, then one quadratic” is the universal skeleton of closed-form TDoA; the Section 6 method shares that skeleton and differs in its coordinate transformation, explicit scalar coefficients, and the completeness of its solvability and accuracy analysis.

6. The Complete Analytical Solution (Hubáček, Veselý & Olivová, 2022)

This section presents, in full detail, the closed-form method of [1] for the sufficiently determined 2D case — three time-synchronized sensors S_1, S_2, S_3 at known positions, one emitter T , line-of-sight propagation — and its extension to overdetermined networks. The authors call the underlying measurement scheme LE-TDOA (*leading-edge* TDoA: each station time-stamps the arrival of the pulse edge against the shared clock, and the differences are formed centrally). A note on symbols: following [1], within this section the propagation speed is written c_l , because the plain letter c is reserved for a transformed sensor coordinate. The measurement model is the one from Section 2:

$$\text{TOA}_i = t_0 + \frac{\sqrt{(x_i - x_t)^2 + (y_i - y_t)^2}}{c_l}, \quad i = 1, 2, 3.$$

6.1 Step 1 — the coordinate transformation, and why it linearizes

Generally the sensors sit anywhere in the plane. The first — and decisive — move is to translate and rotate the coordinate frame $\{x, y\}$ into a frame $\{x^*, y^*\}$ adapted to the sensors: the reference (central) sensor S_1 is moved to the **origin**, and S_2 is rotated onto the **positive x^* -axis**. With $\alpha = \text{atan2}(y_2 - y_1, x_2 - x_1)$, any point (x, y) maps to

$$x^* = (x - x_1) \cos \alpha + (y - y_1) \sin \alpha, \quad y^* = -(x - x_1) \sin \alpha + (y - y_1) \cos \alpha \quad (1)$$

so that the sensors acquire the canonical coordinates $S_1^* = [0, 0]$, $S_2^* = [a, 0]$, $S_3^* = [b, c]$, where a is the S_1 – S_2 baseline length and (b, c) are the transformed coordinates of S_3 . The target becomes $T^* = [x_t^*, y_t^*]$, to be found.

Why does this linearize anything? Write K for the distance from the target to the central sensor, $K = r_1 = c_l(\text{TOA}_1 - t_0)$. Because S_1 now sits at the origin, the target's squared distance to it is simply $K^2 = x_t^{*2} + y_t^{*2}$. When the range equations to S_2 and S_3 are squared (Step 3 below), each produces the quadratic combination $x_t^{*2} + y_t^{*2}$ plus terms linear in x_t^*, y_t^* — and that quadratic combination is *exactly* K^2 , which cancels against the K^2 arising on the measurement side. What survives is **linear in (x_t^*, y_t^*) for any given K** . Moreover, because S_2 lies on the axis, its equation involves only x_t^* (coefficient $-2a$); S_3 contributes both coordinates (coefficients $-2b$ and $-2c$). The transformation thus turns two nonlinear equations into a triangular linear system in (x_t^*, y_t^*) parametrized by the single scalar K — and the entire nonlinearity of TDoA collapses into one scalar quadratic for K . One practical aside: $\cos \alpha = (x_2 - x_1)/a$ and $\sin \alpha = (y_2 - y_1)/a$ are plain coordinate ratios, so no trigonometric function ever needs to be evaluated, and for a fixed sensor network the whole transformation is precomputed once.

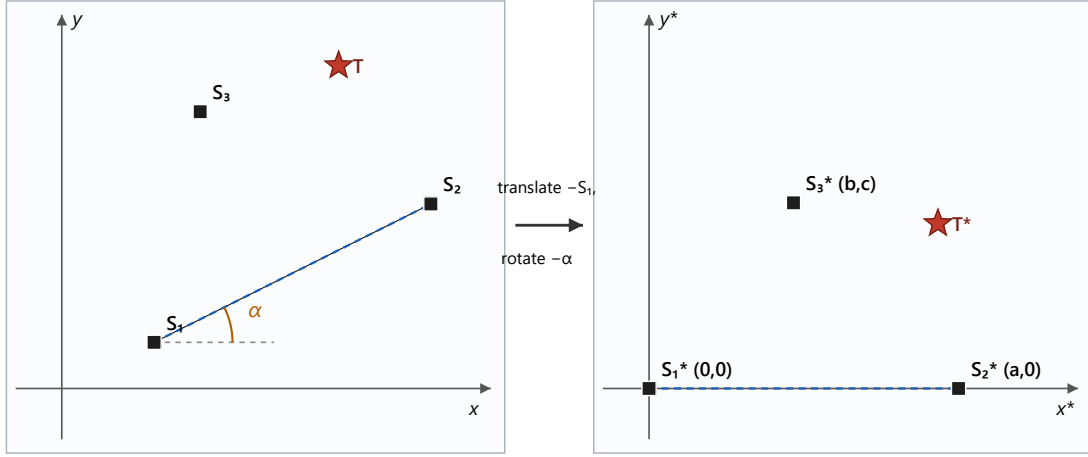


Figure 3. The coordinate transformation of Eq. (1), drawn exactly for sensors $S_1 = (2, 1)$, $S_2 = (8, 4)$, $S_3 = (3, 6)$ and a target $T = (6, 7)$. *Left:* the original frame $\{x, y\}$; $\alpha \approx 26.57^\circ$ is the angle of the $S_1 \rightarrow S_2$ baseline. *Right:* after translating by $-S_1$ and rotating by $-\alpha$, the frame $\{x^*, y^*\}$ has S_1^* at the origin and $S_2^* = (a, 0)$ on the axis, with $a = 45 \approx 6.708$, $S_3^* \approx (3.130, 4.025)$, $T^* \approx (6.261, 3.578)$. Distances are preserved, so ranges and TDOAs are unchanged.

6.2 Step 2 — the measured range differences

With S_1 as reference, the two measured TDOAs are scaled into range differences, named L and R in [1]:

$$L = c_l (\text{TOA}_2 - \text{TOA}_1), \quad R = c_l (\text{TOA}_3 - \text{TOA}_1) \quad (2)$$

and in the transformed frame the geometry ties them to the unknowns through

$$L = \sqrt{(a - x_t^*)^2 + y_t^{*2}} - K, \quad R = \sqrt{(b - x_t^*)^2 + (c - y_t^*)^2} - K, \quad K^2 = x_t^{*2} + y_t^{*2} \quad (3)$$

6.3 Step 3 — algebraic reduction: coordinates affine in K

Move K to the left in (3) and square. For the S_2 equation: $(L + K)^2 = a^2 - 2ax_t^* + (x_t^{*2} + y_t^{*2}) = a^2 - 2ax_t^* + K^2$; the K^2 cancels and solving for x_t^* gives an expression affine in K :

$$x_t^* = A + BK, \quad A = \frac{a^2 - L^2}{2a}, \quad B = -\frac{L}{a} \quad (4)$$

For the S_3 equation: $(R + K)^2 = b^2 + c^2 - 2bx_t^* - 2cy_t^* + K^2$; again K^2 cancels, and substituting $x_t^* = A + BK$ from (4) yields

$$y_t^* = C + DK, \quad C = \frac{b^2 + c^2 - 2bA - R^2}{2c}, \quad D = \frac{-R - bB}{c} \quad (5)$$

A verified correction to the printed paper. The version of record of [1] prints the numerator of C as $b^2 + c^2 - 2aA - R^2$ (with $2aA$). This is a misprint: the coefficient of x_t^* in the squared S_3 equation is $-2b$, as the derivation above makes explicit, so the correct term is **2bA**. The corrected formula was verified numerically for this document — with $2bA$ the solver reproduces ground-truth emitter positions exactly on the example geometry and on eight sign-variant geometries; with $2aA$ it does not. All formulas and numbers in this section use the corrected coefficient. (For geometries where $b = 0$, such as the worked example below, the two versions coincide, which is presumably why the misprint could survive checking.)

6.4 Step 4 — the quadratic in K : candidate ranges from the central sensor

Equations (4) and (5) give the target coordinates as soon as K is known — but K must be consistent with its own definition $K^2 = x_t^{*2} + y_t^{*2}$. Substituting (4)–(5) into that identity: $K^2 = (A + BK)^2 + (C + DK)^2$, which rearranges into a scalar quadratic,

$$MK^2 + 2NK + P = 0, \quad M = B^2 + D^2 - 1, \quad N = AB + CD, \quad P = A^2 + C^2 \quad (6)$$

$$K_{1,2} = \frac{-N \pm \sqrt{N^2 - MP}}{M} \quad (7)$$

(Equation (15) of [1] writes the same quadratic as $MK^2 + NK + P = 0$ with $N = 2(AB + CD)$ and roots $(-N \pm \sqrt{N^2 - 4MP})/(2M)$; the two forms are algebraically identical, and the self-consistent reduced form above is used throughout this document.) The interpretation of (7) is the heart of the method: **the two roots $K_{1,2}$ are the candidate ranges from the emitter to the central sensor S_1** . Every nonlinear aspect of hyperbolic intersection has been condensed into the choice between (at most) two positive numbers. Note also that $P = A^2 + C^2 \geq 0$ always, a small fact with large consequences in Section 6.7.

6.5 Root classification: exactly four cases

Following §3.1 of [1], the root pair (K_1, K_2) falls into exactly one of four classes:

- **Two distinct positive real roots** — the data are consistent with *two* distinct emitter positions: the true target and a *ghost*. This is the inherent ambiguity of exactly determined TDoA (Section 7).
- **A double positive root** (zero discriminant, $N^2 = MP$) — the tangency case: the two hyperbola branches touch rather than cross, the target and its would-be ghost coincide, and the position is unique.
- **Complex roots** (negative discriminant) — no real emitter position is consistent with the data. For noiseless measurements generated by a real emitter this cannot occur (the true $K = r_1$ is always a real root); in practice it signals measurement noise or faulty data, and the fix should be rejected.
- **One positive and one negative root** — the unique-solution case. K is a distance; a negative root is physically meaningless and is discarded, leaving exactly one real target.

6.6 Step 5 — back-substitution and the inverse transformation

Each admissible (positive real) root K is pushed back through (4)–(5) to give $(x_t^*, y_t^*) = (A + BK, C + DK)$, and the transformed coordinates are returned to the original frame by the inverse of (1) — rotate by $+\alpha$, then translate by $+S_1$:

$$x_t = x_t^* \cos \alpha - y_t^* \sin \alpha + x_1, \quad y_t = x_t^* \sin \alpha + y_t^* \cos \alpha + y_1 \quad (8)$$

The full pipeline per fix is therefore: measure $\text{TOA}_{1..3} \rightarrow$ form L, R by (2) $\rightarrow$ evaluate A, B, C, D by (4)–(5) and M, N, P by (6) $\rightarrow$ take the roots (7) $\rightarrow$ classify them (6.5) $\rightarrow$ back-substitute and invert the transformation (8). For a static sensor network, α, a, b, c are precomputed; the per-measurement work is a handful of multiplications and additions and a *single square root*, the one in (7).

6.7 Solvability: the conditions $a = 0, c = 0, M = 0$, and the structure of the plane

The algorithm fails to produce a solution in exactly three situations, each with a clean physical meaning [1]:

- $a = 0$. The denominator of A and B vanishes: S_2 coincides with S_1 , so the network has only two distinct stations and one independent TDOA — a single hyperbola branch, not a fix. No algorithm can rescue this; it is an information deficit, not a formula defect.
- $c = 0$. The denominator of C and D vanishes: S_3 lies on the x^* -axis, i.e. the *three sensors are collinear*. A collinear array measures only distances along, and offsets from, its own line; it cannot distinguish a target from its mirror image across that line. A modified algorithm solves the collinear case up to that unavoidable mirror ambiguity, but the configuration is best avoided by design (Section 10).
- $M = 0$, i.e. $B^2 + D^2 = 1$. The quadratic (6) degenerates to a linear equation with a single finite root; the “second solution” has escaped to infinity. Physically, a target generating data with $M = 0$ would lie at *infinite distance* from S_1 : as the emitter recedes in direction θ , one finds $B \rightarrow \cos \theta$ and $D \rightarrow \sin \theta$, so $M \rightarrow 0$ exactly. The condition can be written as a quadratic relation $R = f(L)$ (Eqs. (18)–(19) of [1]) traced over the admissible interval $L \in (-a, a)$ — the hyperbolic-plane image of “the target is at infinity.”

The sign of M organizes the entire ambiguity structure of the plane, and the organization can be read directly off the quadratic. The product of the roots of (6) is P/M with $P \geq 0$ always; hence if $M < 0$ the roots have opposite signs — one positive, one (discarded) negative — and the fix is *unique*. If $M > 0$ the roots share a sign, and since the true range $K = r_1 > 0$ is always one of them, *both are positive*: the fix is *ambiguous*. Mapping this classification over the $\{x, y\}$ plane (Figs. 3–4 of [1], and Figure 4 below, computed exactly for the example geometry) yields the three-area structure:

- **Area 1** (where $M < 0$): one positive and one negative root — a unique fix. In the example geometry this includes the interior of the sensor triangle and a broad wedge beyond it.
- **Areas 2 and 3** (where $M > 0$): both roots positive — the data admit two real positions. The true target lies in one of the two areas and its ghost in the other (“the first target lies in Area 2 and the second in Area 3, or vice versa” [1]). The two areas are paired: in this document’s convention, a target in Area 3 has its ghost nearer to S_1 than itself (ghost in Area 2), and conversely.
- The $M = 0$ **curve** is the border across which the second solution escapes to infinity — the boundary of the ambiguous zone, where Area 1 meets Areas 2/3. The **zero-discriminant locus** $N^2 = MP$ (tangency, the double-root case of 6.5) lies inside the ambiguous zone and is the seam along which Areas 2 and 3 meet: crossing it, the ghost passes through the target itself. Both loci reappear verbatim in the accuracy analysis: the error of the method blows up exactly where $M \rightarrow 0$ (§3.2 of [1] and Section 8 below).

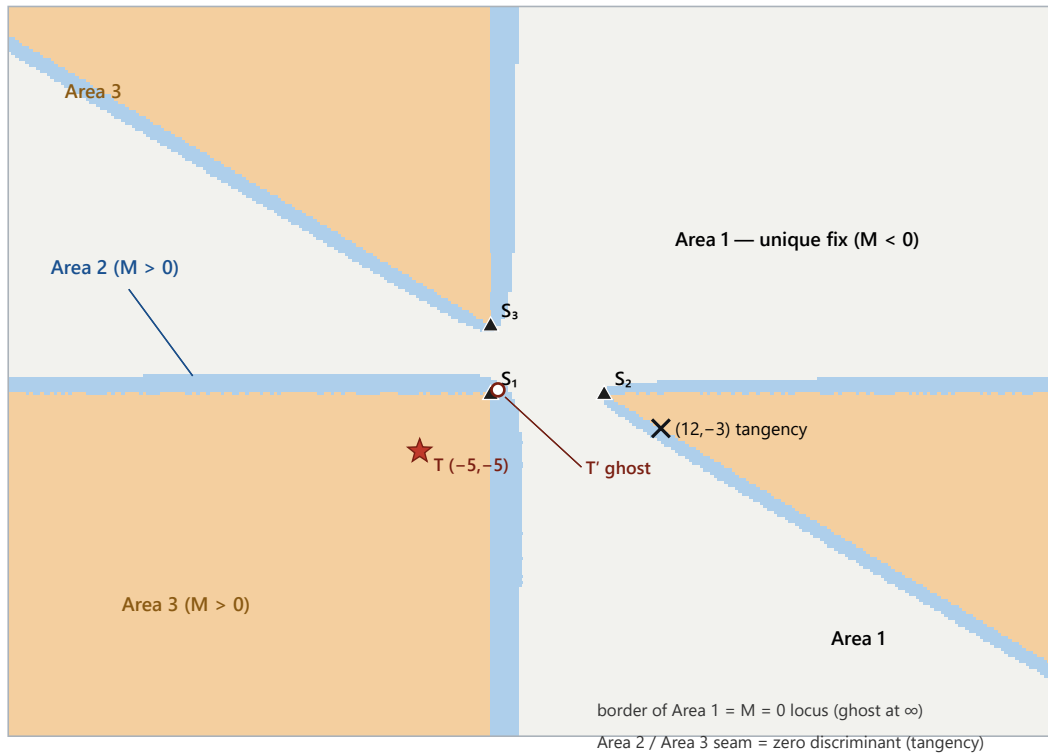


Figure 4. Ambiguity structure of the plane for the example network $S_1 = (0, 0)$, $S_2 = (8, 0)$, $S_3 = (0, 6)$, computed point-by-point from the exact root classification of Eq. (7) (no sketching). Light gray: Area 1, unique fix ($M < 0$: one positive, one negative root). Blue: Area 2 and orange: Area 3, the paired ambiguous areas ($M > 0$: two positive roots); a true target in one has its ghost in the other. The Area 1 boundary is the $M = 0$ locus (ghost at infinity); the blue–orange seam is the zero-discriminant (tangency) locus. The worked example of Table 1 is marked: true target $T = (-5, -5)$ in Area 3, ghost $T' = (0.515, 0.375)$ in Area 2; the $\times$ at $(12, -3)$ is the tangency example, lying exactly on the seam.

6.8 A worked numerical example (independently verified)

Consider the network $S_1 = (0, 0)$, $S_2 = (8, 0)$, $S_3 = (0, 6)$ in arbitrary consistent units (kilometres, if preferred). This network is already in canonical position — S_1 at the origin, S_2 on the x -axis — so $\alpha = 0$, the transformation (1) is the identity, and $a = 8$, $b = 0$, $c = 6$. Let the true emitter be at $T = (-5, -5)$. Table 1 traces every quantity of the algorithm; each line can be reproduced with a hand calculator.

Table 1. The complete analytical solution applied to $S_1 = (0, 0)$, $S_2 = (8, 0)$, $S_3 = (0, 6)$, emitter $T = (-5, -5)$. All values rounded to six decimals; the algebra was verified to machine precision.

Step	Quantity	Value
true ranges	$r_1 = K_{\text{true}}, r_2, r_3$	$\sqrt{50} = 7.071068$, $\sqrt{194} = 13.928388$, $\sqrt{146} = 12.083046$
Eq. (2)	$L = r_2 - r_1, R = r_3 - r_1$	6.857320, 5.011978
Eq. (4)	A, B	1.061072, -0.857165
Eq. (5)	C, D	0.906673, -0.835330
Eq. (6)	M, N, P	0.432508, -1.666885, 1.947930
Eq. (7)	discriminant $N^2 - MP$	1.936011 ($\sqrt{\cdot} = 1.391406$)
Eq. (7)	roots K_1, K_2	7.071068, 0.636934 — both positive: ambiguous
Eqs. (4),(5),(8)	position from K_1	(-5.000000, -5.000000) — the true target
Eqs. (4),(5),(8)	position from K_2	(0.515114, 0.374623) — the ghost

Both roots are positive, so the algorithm honestly reports two candidate emitters: the true target at $(-5, -5)$, at range $K_1 = 7.071068$ from S_1 , and a ghost at $(0.515114, 0.374623)$, at range $K_2 = 0.636934$. Both points reproduce the measured L and R exactly — they are genuinely indistinguishable from three TOAs alone (Figure 5 shows both hyperbola branches passing through both points). Now move the emitter to $T = (12, -3)$. The same pipeline gives $L = -7.369317$, $R = 2.630683$, $A = 0.605823$, $B = 0.921165$, $C = 2.423292$, $D = -0.438447$, $M = 0.040780$, $N = -0.504423$, $P = 6.239367$ — and the discriminant $N^2 - MP = 0$ to machine precision: a **double root** $K = -N/M = 12.369317 = \sqrt{153}$, the tangency case of Section 6.5. The two hyperbola branches touch at $(12, -3)$ instead of crossing, the ghost has merged with the target, and the fix is unique. This emitter lies exactly on the zero-discriminant seam of Figure 4. (The original paper illustrates ambiguity with a kilometre-scale three-sensor example of its own; the example above, chosen for this document, has the advantage that every digit can be checked by hand.)

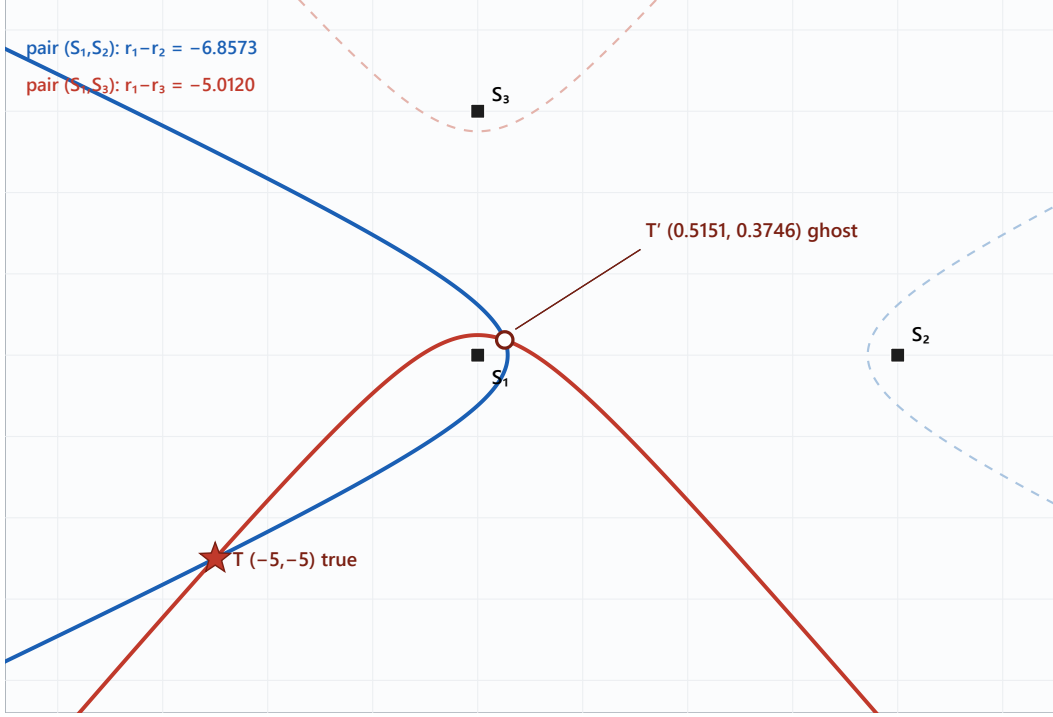


Figure 5. The worked example of Table 1 drawn exactly. Sensors $S_1 = (0, 0)$, $S_2 = (8, 0)$, $S_3 = (0, 6)$; emitter $T = (-5, -5)$. The solid blue curve is the valid branch for the pair (S_1, S_2) ($r_1 - r_2 = -6.857320$, wrapping the nearer sensor S_1); the solid red curve is the valid branch for (S_1, S_3) ($r_1 - r_3 = -5.011978$). The excluded mirror branches are dashed. The two valid branches intersect *twice*: at the true target T and at the ghost $T' = (0.515114, 0.374623)$ near S_1 — the two positive roots K_1, K_2 of Eq. (7) made visible.

6.9 Overdetermined networks: closed form per triple, covariance-weighted fusion

With $N > 3$ sensors in 2D ($N > 4$ in 3D) the system is overdetermined, and [1] extends the analytical solution by combinatorial decomposition rather than by abandoning the closed form. From N sensors one builds all

$$M_{\Delta} = \binom{N}{3}$$

sufficiently determined sub-systems (sensor triples) $TS_1, \dots, TS_{M_{\Delta}}$. Each triple is solved by the closed form of Sections 6.1–6.6, producing a position estimate $\mathbf{T}_i$, and each estimate is given an analytic covariance matrix $\mathbf{C}_i$ by propagating the TOA noise through the algorithm’s chain of substitutions: $\mathbf{C}_i = \mathbf{J} \mathbf{C}_p \mathbf{J}^T$, where $\mathbf{C}_p = \sigma_{\text{TOA}}^2 \mathbf{I}$ is the covariance of the independently measured TOAs and $\mathbf{J} = \partial(x_i, y_i) / \partial(\text{TOA}_1, \text{TOA}_2, \text{TOA}_3)$ is the Jacobian, computed analytically through A, B, C, D, M, N, P (Appendix A of [1] differentiates the entire pipeline in closed form — no numerics required). The fused estimate is the covariance-weighted mean (Eq. (24) of [1]):

$$\mathbf{T} = \left[\sum_i \mathbf{T}_i \mathbf{C}_i^{-1} \right] \left[\sum_i \mathbf{C}_i^{-1} \right]^{-1} \quad (9)$$

which weights each triple’s answer by its information content — accurate triples dominate, poorly conditioned ones are discounted. The fused covariance must account for the fact that the triples are *not* independent (they share sensors, hence share TOA errors), which [1] handles with a correction factor k_e (their Eqs. (25)–(26)):

$$\mathbf{C}_e = k_e \left[\sum_i \mathbf{C}_i^{-1} \right]^{-1}, \quad k_e = \frac{M_{\Delta} r}{s} \quad (10)$$

where M_Δ is the number of triples, r the number of mutually dependent (shared) sensors between sub-systems, and $s = 3$ the minimum number of sensors per sub-system; $k_e = 1$ would correspond to fully independent sub-systems. In the paper's four-sensor simulation ($N = 4$: $M_\Delta = 4$ triples, any two of which share $r = 2$ sensors), $k_e = 4 \cdot 2/3 \approx 2.66$; with sensors at $(0, 0)$, $(-7000, 7000)$, $(7000, 7000)$, $(0, -10000)$ m, a target at $(10000, 50000)$ m and TOA noise $\sigma_{\text{TOA}} = 10$ ns, the fused estimate came out at $(9984, 49940)$ m — a miss distance of $\Delta R = 62.1$ m at a range of roughly 51 km [1].

Finally, the computational argument. Because the per-triple solution is direct, the operation count per fix is fixed and known in advance — a property iterative methods cannot offer — and the comparison study of the 3D extension [13] found that **this analytical route needs approximately 26% fewer mathematical operations than the matrix-form solutions** of the Bard–Ham–Jones type, eliminates matrix inversion (with its conditioning hazards) from the inner loop, is free of trigonometric evaluations, and requires **only one square root** per triple — the discriminant root in (7). For ELINT systems that must compute fixes at the pulse repetition rate of the intercepted emitter, thousands of times per second, this arithmetic economy is the difference between feasible and not [1].

7. Ambiguity: Where the Ghost Lives and How to Exorcise It

The two positive roots of Eq. (7) are not an artifact of the algebra — they report a geometric fact. Two hyperbola branches in the plane can intersect in **two points**, and when they do, both intersections generate *identical* TDOA measurements: the map from position to TDOA pair is one-to-one, but its inverse is not. The forward map sends both T and T' of Figure 5 to the same (L, R) ; no measurement precision at the same three sensors can separate them. Section 6.7 characterized exactly where this happens: a unique-fix region (Area 1, $M < 0$, spurious root negative) and a paired ambiguous zone (Areas 2/3, $M > 0$), bounded by the $M = 0$ locus along which the ghost recedes to infinity and seamed by the tangency locus where ghost and target coincide. The qualitative picture of Figure 4 is worth internalizing: *fixes near and inside the array tend to be unique; ambiguity is a phenomenon of the array's exterior*, and a far emitter pairs with a phantom in a specific conjugate region — often uncomfortably close to the array, like T' in the worked example, only 0.637 units from S_1 while the true emitter is 7.07 units away on the opposite side.

Three remedies exist, in increasing order of robustness. First, **prior constraints**: if the surveillance sector, terrain mask, or maximum plausible range excludes one candidate, the other wins — cheap, but brittle near region boundaries. Second, **a fourth sensor** (fifth in 3D): the additional independent hyperbola passes through the true emitter by construction but generically misses the ghost; the candidate consistent with all measurements is the target. This converts the problem to an overdetermined one in which the ambiguity generically vanishes [1]. Third, equivalently, **consistency across sensor triples** in the fusion framework of Section 6.9: true positions cluster tightly across triples while each triple's ghost lands somewhere else, so selecting per triple the root corroborated by the others (then fusing by Eq. (9)) resolves the ambiguity with machinery already in place. Iterative methods, by contrast, never even enumerate the second solution — near ambiguous-area borders an unlucky initialization simply converges to the wrong one [1, 2]; the analytical solution's ability to list and classify *all* solutions is what makes systematic disambiguation possible.

8. Errors, the Cramér–Rao Bound, and GDOP

8.1 From TOA noise to TDOA covariance in two lines

Model each measured arrival time as truth plus independent, identically distributed noise: $\widehat{\text{TOA}}_i = \text{TOA}_i + n_i$, with $\mathbb{E}[n_i] = 0$, $\mathbb{E}[n_i n_j] = \sigma_i^2 \delta_{ij}$. The common-reference range-difference measurements are $d_i = c(\widehat{\text{TOA}}_{i+1} - \widehat{\text{TOA}}_1)$, $i = 1, \dots, N-1$, with errors $c(n_{i+1} - n_1)$. The covariance is immediate:

$$\mathbb{E}[c^2(n_{i+1} - n_1)(n_{j+1} - n_1)] = c^2(\mathbb{E}[n_{i+1} n_{j+1}] + \mathbb{E}[n_1^2]) = c^2 \sigma_i^2 (\delta_{ij} + 1),$$

$$\boxed{\mathbf{R} = c^2 \sigma_i^2 (\mathbf{I} + \mathbf{1}\mathbf{1}^\top)} \quad \text{i.e.} \quad \text{Var} = 2c^2 \sigma_i^2, \quad \text{Cov} = c^2 \sigma_i^2.$$

Every common-reference TDOA contains the same reference-sensor error n_1 , so the measurements are *correlated*, with variance $2c^2\sigma_t^2$ on the diagonal and covariance $c^2\sigma_t^2$ everywhere else. Ignoring this correlation (treating $\mathbf{R}$ as diagonal) is a classic implementation error that degrades any weighted estimator. A useful identity: $(\mathbf{I} + \mathbf{1}\mathbf{1}^\top)^{-1} = \mathbf{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$ for the $(N - 1)$ -dimensional TDOA vector.

8.2 Geometry enters: the Jacobian and the CRLB

Linearizing the measurement map about the true position, the row of the Jacobian $\mathbf{H}$ for measurement $d_i = r_{i+1} - r_1$ is the gradient

$$\nabla_{\mathbf{x}} (r_{i+1} - r_1) = \mathbf{u}_{i+1} - \mathbf{u}_1, \quad \mathbf{u}_i = \frac{\mathbf{x} - \mathbf{s}_i}{r_i}$$

where $\mathbf{u}_i$ is the unit vector pointing from sensor i toward the emitter. Each row of $\mathbf{H}$ is thus a *difference of unit vectors* — small when the emitter sees two sensors in nearly the same direction (far field), large when the directions differ strongly (emitter amid the array). For unbiased estimation with Gaussian TDOA errors, the Cramér–Rao lower bound on the position covariance is

$$\mathbf{P} = (\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1} = c^2\sigma_t^2 \left(\mathbf{H}^\top (\mathbf{I} + \mathbf{1}\mathbf{1}^\top)^{-1} \mathbf{H} \right)^{-1}.$$

The scalar noise level $c^2\sigma_t^2$ factors out completely, leaving a purely geometric core. This expression is the classical result of Torrieri’s statistical theory of passive location [9], and for the exactly determined three-sensor case it coincides with the analytic covariance $\mathbf{J} \mathbf{C}_p \mathbf{J}^\top$ of Section 6.9 — the closed-form algorithm is an exact inverse of the measurement map, so propagating TOA noise through it reproduces the bound. A reassuring structural fact: $\mathbf{P}$ is **invariant to the choice of reference sensor**. Changing reference replaces the TDOA vector $\mathbf{d}$ by $\mathbf{T}\mathbf{d}$ for an invertible matrix $\mathbf{T}$, whence $\mathbf{H} \rightarrow \mathbf{T}\mathbf{H}$, $\mathbf{R} \rightarrow \mathbf{T}\mathbf{R}\mathbf{T}^\top$, and $\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H}$ is unchanged. No sensor is statistically privileged by being the reference.

8.3 GDOP: a convention, stated explicitly

The **geometric dilution of precision** condenses $\mathbf{P}$ into one number. The convention used throughout this document — conventions differ across the literature, so it must be stated — is

$$\text{GDOP} \equiv \frac{\sqrt{\text{tr} \mathbf{P}}}{c \sigma_t} = \sqrt{\text{tr} \left[(\mathbf{H}^\top (\mathbf{I} + \mathbf{1}\mathbf{1}^\top)^{-1} \mathbf{H})^{-1} \right]}$$

i.e. the root-sum-square position error per unit of range-equivalent timing error. Because $c\sigma_t$ cancels, **GDOP is purely geometric**: it depends only on where the sensors and the emitter are, not on how noisy the clocks are. The achievable accuracy then factorizes memorably as RMS position error = GDOP $\times$ $c\sigma_t$: geometry times measurement quality. A GDOP of 1.5 with 10 ns timing (3 m range-equivalent) means roughly 4.5 m RMS error; the same hardware in a GDOP-200 region delivers 600 m. Figure 6 maps GDOP exactly for the example triangle: values are modest (1–2) inside and near the array, grow through single digits in the surrounding annulus, and blow up along the lines through sensor pairs and toward the far field.

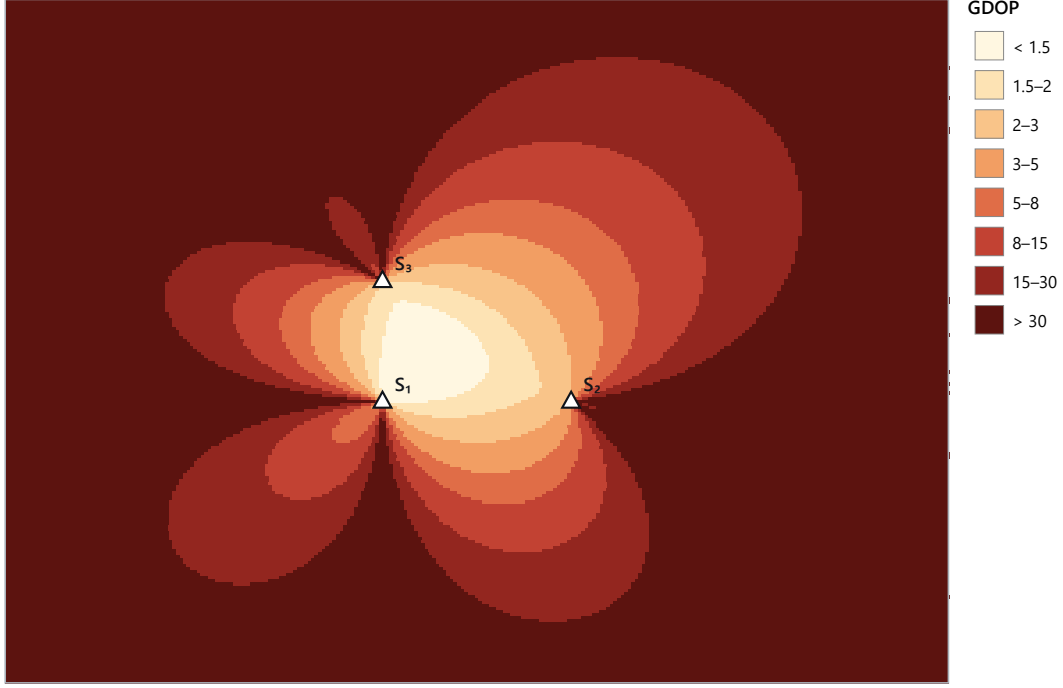


Figure 6. GDOP computed exactly on a grid for the array $S_1 = (0,0)$, $S_2 = (8,0)$, $S_3 = (0,6)$ (white triangles), using $\text{GDOP} = \text{tr}[(\mathbf{H}^\top \mathbf{W}^{-1} \mathbf{H})^{-1}]$ with $\mathbf{W} = \mathbf{I} + \mathbf{1}\mathbf{1}^\top$. Best accuracy (light) occurs inside and immediately around the triangle; GDOP deteriorates outward roughly quadratically with range and diverges along the sensor-pair lines, where the unit-vector differences become collinear. Compare the dark lobes with the $M = 0$ blow-up loci of Figure 4 — the accuracy map and the solvability map degrade along the same directions, as the paper’s “accuracy map” demonstrates at scale [1].

8.4 Error ellipses

The covariance $\mathbf{P}$ is rendered geometrically as an **error ellipse**. Eigendecomposition $\mathbf{P} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$ gives the principal directions (columns of $\mathbf{V}$) and the 1σ semi-axes $\sqrt{\lambda_1}, \sqrt{\lambda_2}$. One subtlety bites newcomers: in two dimensions the “ 1σ ellipse” contains not 68% but $1 - e^{-1/2} = 39.35\%$ of the probability mass; a 95% region requires scaling the 1σ axes by $\sqrt{\chi_{2,0.95}^2} = 2.4477$. Hubáček et al. plot the major-axis length of these ellipses over the plane on a log scale as their “accuracy map” (computed with $\sigma_{\text{TOA}} = 10$ ns), noting that the ellipses are highly eccentric in the far field — range error dwarfs cross-range error — and that the blow-up loci coincide with the $M = 0$ curves of the solvability analysis [1]. Torrieri [9] remains the standard reference for this error-analysis toolkit, and Van Trees [12] for the underlying estimation theory.

8.5 Far-field degeneration: TDoA becomes AoA

Let the emitter recede to ranges much larger than the array aperture. Each hyperbola branch hugs its asymptote ever more closely, and the measurement ceases to distinguish points along the asymptote — it measures only the *direction*. In the limit, a compact TDoA array degenerates into an angle-of-arrival (AoA) sensor: the TDOA across a baseline $\mathbf{b}$ tends to $\Delta r \rightarrow -\mathbf{b} \cdot \hat{\mathbf{u}}$, the projection of the baseline on the arrival direction — an interferometer’s observable. Bearing remains estimable (error growing linearly with range), but **range becomes unobservable**: range information lives in the wavefront curvature across the array, which scales as $(\text{aperture})^2/r$, so range variance — and with it GDOP — blows up quadratically and worse. This is the operational meaning of both the $M = 0$ condition (“target at infinity”, Section 6.7) and the dark far-field lobes of Figure 6, and it is why long-range passive-location systems use very long baselines, fuse TDoA with FDOA or AoA, or settle for bearings-only tracking until geometry improves.

8.6 From waveform to σ_t : the delay-estimation CRLB

Everything above took the timing accuracy σ_t as given. It is itself bounded by the physics of the signal: for delay estimation from a received waveform, the Cramér–Rao bound takes the form

$$\sigma_t \geq \frac{1}{2\pi\beta\sqrt{\text{SNR}}}, \quad \beta^2 = \frac{\int f^2 |S(f)|^2 df}{\int |S(f)|^2 df}$$

where β is the *effective (RMS) bandwidth* of the signal spectrum $S(f)$ and SNR is the effective post-integration signal-to-noise ratio (the two-sensor cross-correlation expressions appear in Knapp & Carter [10]; the single-channel form and derivation in Kay [11]). The bound is the bridge from waveform to geometry: *bandwidth, not power alone, buys timing accuracy*. A 1 MHz-wide signal at healthy SNR supports σ_t in the tens of nanoseconds; a 100 MHz radar pulse supports sub-nanosecond timing; an unmodulated carrier supports almost nothing — its correlation peak is too blunt to locate in time. Multiplying σ_t by c and by the GDOP of Figure 6 chains the whole error budget: spectrum $\rightarrow$ timing $\rightarrow$ geometry $\rightarrow$ position.

9. Estimating the TDOA from Signals

Sections 2–8 assumed the time differences were available; extracting them from raw signals is its own discipline. For two sensors receiving delayed, noisy copies of a common waveform, $x_1(t) = s(t) + n_1(t)$ and $x_2(t) = s(t - \tau) + n_2(t)$, the natural estimator is the lag of the **cross-correlation peak**: $\hat{\tau} = \arg \max_{\tau} \int x_1(t) x_2(t + \tau) dt$. For pulsed signals with sharp edges (the radar case of [1]), leading-edge thresholding against the synchronized clock — the LE-TDOA scheme — can replace correlation entirely, trading some noise performance for simplicity and per-pulse speed.

The workhorse refinement is the **generalized cross-correlation (GCC)** of Knapp and Carter [10]: correlate in the frequency domain after weighting the cross-spectrum $G_{x_1 x_2}(f)$ by a function $\psi(f)$, $R_{x_1 x_2}^{(g)}(\tau) = \int \psi(f) G_{x_1 x_2}(f) e^{j2\pi f \tau} df$. The maximum likelihood weighting emphasizes frequencies of high coherence; the most popular practical choice is **PHAT** (phase transform), $\psi(f) = 1/|G_{x_1 x_2}(f)|$, which discards magnitude and keeps only cross-phase. PHAT whitens the signal so the correlation collapses toward a delta-like spike at the true delay — strikingly robust to reverberation and moderate multipath, which is why it dominates acoustic applications — at the cost of being suboptimal at low SNR, where it amplifies bands containing no signal.

One structural hazard deserves emphasis: **periodic signals make the correlation ambiguous**. If $s(t)$ is periodic with period T_p — a pulse train with constant repetition interval, a CW tone — the cross-correlation has identical peaks at $\hat{\tau} + kT_p$ for every integer k , and the TDOA is determined only modulo T_p . If the geometry permits true delays exceeding T_p (baselines longer than cT_p), the wrong peak can be selected, producing position errors no SNR will fix. Remedies: baselines short enough that $|\Delta t| < T_p/2$ is guaranteed; PRI stagger or jitter, which breaks the periodicity and isolates the true peak; and multi-hypothesis association that carries several candidate delays into localization and lets geometric consistency choose — the same logic that resolves the geometric ghost in Section 7.

10. Practical Considerations

Clock synchronization. The TDoA observable exists only relative to a shared time base, and every nanosecond of inter-sensor clock error converts to 30 cm of range-difference error at light speed. GPS-disciplined oscillators are the field workhorse, holding distributed sensors to a common epoch within tens of nanoseconds (a few, with carrier-phase techniques) — though they fail together under GNSS jamming, which matters in exactly the electronic-warfare scenarios where TDoA is most wanted. Fiber-based time transfer — notably the White Rabbit extension of IEEE 1588 PTP, developed at CERN — achieves sub-nanosecond synchronization over kilometres of fiber and suits fixed installations. Wireless schemes (two-way time transfer, or a reference emitter at a known position whose own TOAs calibrate the network in situ) cover

mobile deployments; the reference-emitter trick elegantly turns synchronization into one more localization problem whose answer is known.

Multipath and NLOS propagation. In urban, indoor and littoral environments the dominant error is usually not clock noise but propagation: a reflected or diffracted path arrives later than the (possibly blocked) direct path, biasing the measured TOA *late* — a systematic, geometry-dependent bias, not zero-mean noise, so it does not average away. Leading-edge detection (earliest plausible arrival, as in LE-TDOA), NLOS identification and down-weighting of suspect sensors, and robust estimators that tolerate outlier TDOAs are the standard defenses; the residual-consistency checks made cheap by closed-form solutions (Section 6.9) double as a practical NLOS alarm.

Sensor geometry. Figures 4 and 6 are design guidance, not just analysis. GDOP is small where the unit vectors to the sensors span directions widely — so, where the mission allows, *surround the target area*: an emitter inside the sensors' convex hull enjoys low GDOP and (typically) an unambiguous fix, while an emitter far outside suffers both inflated error and ghost risk. *Avoid collinearity*: three sensors on a line are the degenerate $c = 0$ case of Section 6.7, with an unresolvable mirror ambiguity across the array line and GDOP divergence along it; near-collinear arrays inherit softened versions of both diseases. Baselines should match the intended operating range — far-field operation pushes the system into the bearing-only regime of Section 8.5 regardless of timing quality.

Propagation-speed uncertainty. The conversion $\Delta r = c \Delta t$ is only as good as c . For RF the speed is effectively exact over tactical ranges. For acoustics it is not: the speed of sound in air varies as $c \approx 331.3 + 0.606 T_{\text{C}}$ m/s — nearly 2% across a 10 °C swing — and wind advects the medium, adding directional bias; underwater, the sound-speed profile refracts paths outright. Acoustic deployments therefore carry thermometers (or estimate c jointly with position as an extra unknown), and serious underwater systems ray-trace through measured profiles.

Data association with multiple emitters. Sections 2–8 assumed all sensors measured *the same signal*. A real environment offers many emitters and many pulses, and the system must decide which detection at sensor 2 corresponds to which at sensor 1 before any hyperbola may be drawn. Mis-association creates *combinatorial ghosts* — spurious intersections from mixed-emitter TDOA sets — distinct from, and additional to, the geometric ghost of Section 7. ELINT practice deinterleaves pulse trains first (clustering by RF, pulse width, PRI), associates across sensors using waveform similarity and feasibility gates (a pairing is impossible unless $|\Delta t| \leq d/c$ — the Section 3 validity condition reused as a filter), then localizes; tracking filters downstream suppress surviving ghosts by their kinematic inconsistency.

11. Summary and Glossary

TDoA localization solves the defining problem of passive surveillance: positioning an emitter that never discloses its transmit time. Differencing synchronized arrival times eliminates the unknown t_0 ; each range difference confines the emitter to one hyperbola branch with foci at the sensor pair; three sensors in 2D (four in 3D) make the system exactly determined, more make it overdetermined. The complete analytical solution of Hubáček, Veselý and Olivová [1] — translate and rotate so the reference sensor sits at the origin and a second sensor on the axis, reduce to coordinates affine in the central range K , solve one quadratic $MK^2 + 2NK + P = 0$ whose roots are the candidate ranges, classify the roots, invert the transformation — delivers every solution explicitly, exposes the ambiguity structure of the plane (Areas 1/2/3, the $M = 0$ and tangency loci), propagates errors analytically, extends to overdetermined networks by covariance-weighted fusion of sensor triples with the $k_e = M_{\Delta r}/s$ dependence correction, and does all this with roughly a quarter fewer operations than matrix-form alternatives and a single square root per fix. Accuracy is governed by the factorization (position error) = GDOP $\times c \sigma_t$: geometry sets GDOP, bandwidth and SNR set σ_t through the delay-estimation CRLB, and the engineering of Section 10 determines whether the theoretical budget is achieved in the field.

Glossary

TOA — Time of Arrival

The absolute time at which a signal reaches a sensor, measured against the system clock; equals the unknown transmit time plus range over propagation speed.

TDOA — Time Difference of Arrival

The difference between the TOAs at two sensors; independent of the transmit time, it scales by c into a range difference and defines a hyperbolic locus.

FDOA — Frequency Difference of Arrival

The difference in received Doppler shift between sensors with relative motion; complements TDOA (notably from moving platforms) by adding loci that resolve range and velocity.

AoA — Angle of Arrival

The direction from which a signal arrives, measured by an array or interferometer; the far-field limit of TDoA across a compact array (Section 8.5).

RSS — Received Signal Strength

Localization observable based on received power and a path-loss model; coarse but cheap, requiring no synchronization.

GDOP — Geometric Dilution of Precision

The purely geometric amplification factor from range-equivalent measurement error to position error; here $\text{GDOP} = \sqrt{\text{tr } \mathbf{P}} / (c\sigma_t)$.

CRLB — Cramér–Rao Lower Bound

The information-theoretic floor on the covariance of any unbiased estimator; used here both for position (given TDOA noise) and for delay (given waveform and SNR).

Multilateration

Position fixing from intersecting loci generated by range or range-difference measurements to multiple known stations; hyperbolic multilateration is the TDoA case.

Ghost / ambiguity

A second emitter position producing measurements identical to the true one — geometrically, the second intersection of hyperbola branches; algebraically, the second positive root of the quadratic (7). Distinct from combinatorial ghosts caused by mis-association of detections (Section 10).

12. References

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